Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/05/2020
PI Name:	Wu, Honglu Ph.D.		
Project Title:	DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-483-6470
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Human Adaptation and Countermeasures Office		
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Zip Code:	77058-3607	**Congressional District:**	22
Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-MIXEDTOPICS. Appendix E: Behavioral Health & Human Health Countermeasures Topics
Start Date:	01/01/2016	**End Date:**	09/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2023 per NASA JSC. (Ed., 11/18/21) NOTE: End date changed to 9/30/2022 per PI (Ed., 10/6/20) NOTE: Extended to 3/31/2021 per PI (March 2019)--Ed., 10/3/19		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feiveson, Alan Ph.D. (NASA Johnson Space Center) Zhang, Ye Ph.D. (NASA Kennedy Space Center) Moreno-Villanueva, Maria Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	NOTE: This is an integrated project consisting of Dr. Brian Crucian's "Functional Immune Alterations, Latent Herpesvirus Reactivation, Physiological Stress, and Clinical Incidence Onboard the International Space Station" directed research; and Dr. Richard Simpson's "The Impact of an ISS Mission on the Anti-Viral and Functional Properties of NK-cells, T-cells, B-cells and Dendritic Cells," Dr. Kanokporn Rithidech's "Effects of Space Flights on the Proteome of Astronauts' Plasma," and Dr. Honglu Wu's "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" solicited research. The unique radiation environment encountered in space has been known to cause deleterious effects in humans, and these effects are of special concern for prolonged space missions beyond the protective terrestrial magnetosphere. DNA lesions, namely single- and double-strand breaks, are the most common of these effects and the most likely to compromise health. In addition to radiation, astronauts in space experience microgravity and other stress factors. In human cells, immune cells in particular, stress-induced catecholamine and corticoid release, as well as increased levels of inflammation, are known to induce DNA damage and impair DNA repair, which in turn results in single and double-strand breaks. Intact DNA repair machinery is crucial for immune cell functionality. Therefore understanding the interaction between DNA damage response and immune response is an important step towards improving knowledge of the mechanisms associated with immune dysfunction in the space environment. In this study, we propose to quantify DNA damage in astronauts' lymphocytes by measuring the frequency of single- and double-strand breaks. We will also perform RNA-seq analysis with a focus on pathways involved in DNA damage response and/or inflammatory response. Measures of DNA damage obtained before, during, and after long-duration International Space Station (ISS) missions will then be compared with the mRNA levels. The outcomes in the present study will also be compared to the measured T cell function, cytokine profiles and viral reactivation in the Integrated Immune Flight Study, which is the parental project of our proposed study.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Sample collection. To date, collection of blood samples has been performed on 10 ISS crewmembers and 10 matching ground controls at all of the 7 (6 for ground controls) time points (L-180, L-45, mid-flight, late-flight, R+0, R+30, and R+90). Collection from the 11th crewmember and the ground control has not begun. Sample analysis RNA-seq analysis. Gene expression sample analysis using the RNA-seq technique was completed for 8 ISS crewmembers and 8 matching ground controls. MicroRNA analysis. MicroRNA sample analysis using the RNA-seq technique was completed for 8 ISS crewmembers and 8 matching ground controls. PCR analysis. Analysis of expression of DNA damage and immune function related genes using the PCR-array technique was completed for 4 ISS crewmembers and 4 matching ground controls. Supplies to analyse the next set of 4 crewmembers and ground controls have been purchased and received. DNA damage analysis. DNA damage analysis is being performed at University of Konstanz, Germany. Data analysis RNA-seq data. Analysis of RNA-seq data was completed for 4 ISS crewmembers and 4 matching ground controls. Initial bioinformatics analysis of the next set of 4 crewmembers and 4 matching ground controls has begun. MicroRNA data. Analysis of microRNA data was completed for 4 ISS crewmembers and 4 matching ground controls.
Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Moreno-Villanueva M, Zhang Y, Feiveson A, Mistretta B, Pan Y, Chatterjee S, Wu W, Clanton R, Nelman-Gonzalez M, Krieger S, Gunaratne P, Crucian B, Wu H. "Single-cell RNA-sequencing identifies activation of TP53 and STAT1 pathways in human T lymphocyte subpopulations in response to ex vivo radiation exposure." Int J Mol Sci. 2019 May 10;20(9):E2316. https://doi.org/10.3390/ijms20092316 ; PMID: 31083348; PMCID: PMC6539494 , May-2019
Articles in Peer-reviewed Journals	Feiveson A, George K, Shavers M, Moreno-Villanueva M, Zhang Y, Babiak-Vazquez A, Crucian B, Semones E, Wu H. "Predicting chromosome damage in astronauts participating in International Space Station missions." Sci Rep. 2021 Mar 5;11(1):5293. https://doi.org/10.1038/s41598-021-84242-5 ; PMID: 33674665; PMCID: PMC7935859 , Mar-2021